HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged system-level tests for pre-silicon environments using the **Incisive simulator** and **Palladium emulator**, contributing to SoC validation.
- Participated in team meetings and discussions to gain insight into industry design team methodologies for building chiplets and complex IC products.
- Automated test execution and analysis processes through **Python scripting**, reducing manual effort for the validation team by approximately **15%**.
- Configured and managed test environments on **AWS EC2** instances, ensuring stability and availability for system-level validation tasks.
- Performed code reviews for validation scripts and test cases, adhering to established coding standards and best practices within the team.
- Analyzed test results and system logs using **Splunk** to identify root causes of failures and debug issues in the pre-silicon environment.
- Containerized validation tools and test harnesses using **Docker** for consistent deployment across different development and testing stages.

SKILLS

Simulation & Emulation – Incisive simulator, Palladium emulator

Programming Languages – Python, Java, SQL

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Scripting & Automation – Shell scripting, Jenkins pipelines

Operating Systems – Linux

Tools & Methodologies – Jira, Confluence, Agile, Git, Docker

PROJECTS

System-Level Test Automation Framework

- Developed a Python-based framework to automate the creation, execution, and analysis of system-level tests for IC product validation.
- Integrated the framework with **Incisive simulator** and **Palladium emulator**, enabling automated test runs and result parsing.
- Implemented logging and reporting mechanisms that provided detailed insights into test outcomes, reducing debugging time by an estimated **20%**.

IC Product Validation Environment Setup

- Configured and maintained a Linux-based test environment on **AWS EC2** for pre-silicon validation activities.
- Scripted environment setup and teardown processes using **shell scripting** and **Jenkins** pipelines for repeatable deployments.
- Managed test data and simulation outputs using **AWS S3**, ensuring efficient storage and retrieval for analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)